SYSC 5906 G

Methodologies for Discrete Event Modeling and Simulation

A Cell-DEVS Based Simulation of Bee Foraging Using Cadmium

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GitHub: https://github.com/GAchuzia/Cell-DEVS-Bee-Foraging

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Defining the Bee Nectar Foraging System

Model Choice Overview

Bee nectar foraging is a natural system in which pollinators interact with spatially distributed flower patches, collecting nectar and facilitating pollen transfer between plants. As pollinators visit flower patches, nectar availability decreases due to consumption, while over time nectar regenerates. At the same time, pollen is transferred between patches through pollinator movement, influencing plant reproduction and resource dynamics.

The goal of this project is to simulate a bee foraging and plant–pollinator interaction system using a Cell-DEVS model. In this model, the environment is represented as a grid where each cell corresponds to a localized nectar patch. Each cell maintains state variables representing nectar levels, pollen availability, and pollinator activity.

The model captures the following:

- Spatial distribution of nectar patches across the environment
- Consumption and regeneration of nectar over time
- Pollen transfer between patches via pollinator movement
- Local interactions between neighboring cells influencing resource dynamics

This model allows for the analysis of how local interactions between pollinators and plant resources lead to emergent patterns such as resource depletion, recovery, and spatial redistribution across the system.

Overview of the Paper

The paper presents a model of plant–pollinator interactions that focuses on how nectar, pollen, and pollinator behavior influence each other over time. It describes how flower patches produce nectar and pollen, which are then consumed and transported by pollinators such as bees.

As pollinators move between patches, they reduce nectar levels through consumption and transfer pollen between plants, contributing to reproduction. At the same time, nectar and pollen regenerate, creating a continuous cycle of resource use and renewal.

The model shows how these interactions create dynamic patterns in the environment, where local changes in one patch can influence neighboring patches through pollinator movement. Overall, the paper provides a solid framework for modelling resource availability and pollinator behavior in a discrete system [1].

Cell Representation

Each cell in the automaton will represent a region called a 'Nectar Patch' with predefined dimensions. The information stored by the nectar patches will include:

- Liquid nectar levels
- Pollen grain levels
- Visitation rates from pollinators

Update Rule for Cellular Automata

At each simulation step, every cell updates its state based on local resource levels and interactions with neighboring cells. Each cell represents a nectar patch containing nectar and pollen resources, as well as pollinator activity.

Nectar levels increase over time through regeneration, with the rate influenced by the amount of pollen available in the cell. Cells with higher pollen levels produce more nectar, reflecting successful pollination. At the same time, nectar is reduced when bees visit the patch, with greater visitation leading to higher consumption.

Pollen levels are updated based on pollinator movement between neighboring cells. When bees move from one cell to another, they transfer pollen, increasing pollen levels in the destination cell while reducing it in the source cell. This creates a spatial interaction where neighboring patches influence each other through pollinator activity.

Bee visitation is influenced by nectar availability. Cells with higher nectar levels attract more pollinators, while cells with low nectar levels lose visitors over time. This results in competition between neighboring patches, where resource-rich cells attract more activity and resource-poor cells become less active.

Simulating the Bee Nectar Foraging System
Design Specification

The nectar–pollinator system is simulated on a two-dimensional grid of size 10×10, where each cell represents a localized nectar patch. Each cell maintains state variables describing nectar levels, pollen availability, and pollinator presence.

Initial Values

● Nectar Level
Each cell is initialized with a fixed amount of nectar. This value may be uniform across the grid or varied depending on the experimental scenario.

● Pollen Level
Each cell is assigned an initial pollen level, representing the pollination state of the patch. This may start at zero or a predefined value depending on the test configuration.

● Pollinator Presence
Each cell contains an initial number of bees, which may be uniformly distributed or concentrated in specific regions of the grid.

● Grid Configuration
The simulation grid size is fixed at 10×10 cells. The configuration can be modified to test different spatial distributions of nectar, pollen, and pollinators.

All initial values are provided through configuration files used to initialize the simulation environment.

Rules

At every simulation step update, the rules will control how the nectar patches share resources and evolve over time. These rules will govern the equilibrium of nectar levels, pollen levels, pollinator presence and grid configuration across their interactions.

Update Nectar Rule

The amount of nectar in a cell after an update depends on the current nectar level, the amount of pollen present in the cell, and the number of pollinators visiting the cell.

Nectar regenerates over time, with higher pollen levels increasing the regeneration rate due to successful pollination. At the same time, nectar is reduced based on the number of bees consuming it.

This is modeled as:

$$N_{0,0} = N_{0,0} + R(P_{0,0}) - C(B_{0,0})$$

Where:
- $N_{0,0}$ is the current nectar level
- $R(P_{0,0})$ is the regeneration function influenced by pollen
- $C(B_{0,0})$ is nectar consumption based on bee activity

The nectar value is bounded by a maximum capacity. If the value exceeds this limit, it is truncated. If nectar reaches zero, the cell becomes unattractive to pollinators.

Update Pollen Rule

The pollen level in a cell is updated based on pollinator movement between neighboring cells. When bees move from one cell to another, they carry pollen with them, increasing the pollen level in the destination cell and decreasing it in the source cell.

This creates a diffusion-like effect where pollen spreads across neighboring cells through pollinator activity.

This is modeled as:

$$P'_{0,0} = P_{0,0} + \sum \text{incoming_transfer} - \sum \text{outgoing_transfer}$$

Where pollen transfer depends on the number of bees moving between neighboring cells.

Pollen values are constrained between minimum and maximum bounds to ensure realistic behavior.

Pollinator Movement Rule

The number of pollinators in each cell is updated based on nectar availability in neighboring cells. Bees are more likely to move toward cells with higher nectar levels and leave cells with low nectar levels.

Movement is restricted to neighboring cells, allowing local interactions to drive system behavior.

If multiple neighboring cells have higher nectar levels, bees distribute themselves proportionally or move toward the highest-valued cell.

This rule creates competition between patches, where resource-rich cells attract more pollinators over time.

Neighbour Interaction Rule

Each cell exchanges information with its neighboring cells to determine pollinator movement and pollen transfer. Neighboring cells influence each other through differences in nectar levels and pollinator presence.

Cells with higher nectar levels attract pollinators from neighboring cells, while pollen is redistributed as bees move across the grid.

This interaction allows local changes in one cell to propagate across the environment, creating spatial patterns in resource distribution.

Resource Constraints Rule

All state variables are bounded to ensure realistic simulation behavior:

- Nectar levels cannot exceed a maximum capacity or fall below zero
- Pollen levels are constrained within defined limits
- Pollinator counts are non-negative

Model Structure and Coupling

This section defines the atomic and coupled models' structures in the Bee Foraging System, with respect to the Cell-DEVS modelling framework. The system consists of one atomic model and one coupled model.

Atomic Model

$$IDC = \langle X,\ Y, S, \theta,\ I, N, d, \tau, \delta_{int}, \delta_{ext}, \lambda, D \rangle$$

X (Input Set)

The input set is received from the neighbouring cells and is represented as a vector containing the following values:

- Nectar levels from neighbors (int)
- Pollen levels from neighbors (int)
- Pollinator movement from neighbors (int)

Y (Output Set)

- Updated nectar level (int)
- Updated pollen level (int)
- Updated pollinator count (int)

S (State Set)

- N = nectar level
- P = pollen level
- B = number of bees

d (delay of the cell)

- 1 time unit

τ (Local Computing Function)

The local transition function τ determines the next state of each cell based on its current state and the states of its neighboring cells.

For each cell (0,0), the updated values are computed as follows:

- $N'_{0,0} = N_{0,0} + R(P_{0,0}) - C(B_{0,0})$
- $P'_{0,0} = P_{0,0} + \sum \text{incoming_pollen} - \sum \text{outgoing_pollen}$
- $B'_{0,0} = B_{0,0} + \sum \text{incoming_bees} - \sum \text{outgoing_bees}$

Where:

- $R(P_{0,0})$ represents nectar regeneration influenced by local pollen levels
- $C(B_{0,0})$ represents nectar consumption based on bee activity
- incoming_pollen and outgoing_pollen depend on pollinator movement between neighboring cells
- incoming_bees and outgoing_bees are determined by relative nectar levels between the current cell and its neighbors

The function enforces the following constraints:

- $0 \leq N'_{0,0} \leq Nmax$
- $0 \leq P'_{0,0} \leq Pmax$
- $B'_{0,0} \geq 0$

Coupled Model

$$\mathbf{GCC} = < \mathbf{Xlist}, \mathbf{Ylist}, I, X, Y, \mathbf{n}, \{\mathbf{t_1},...,\mathbf{t_n}\}, \eta, \mathbf{N}, C, B, Z, \text{select} >$$

- I (Input Interface): $< Px, Py >=< \emptyset, \emptyset >$

- X (Set of Input Values): { nectar, pollen, pollinator movement }
- Y (Set of Output Values): { updated nectar, pollen, pollinators }
- X list (list of external inputs): Ø
- Y list (list of external outputs): Ø
- n (Neighborhood Size): 5 (Von Neumann Neighborhood)
- N (Relative Positions of Neighboring Cells): {(0, 0),(−1, 0),(1, 0),(0, 1),(0,−1)}
- {m, n} (Grid Size {row, column}): {10, 10}
- C (Set of All Cells): $\{C\,ij \,/\, i \in [1, 10], j \in [1, 10]\}$
- B (Border Condition): Ø
- Z (Neighboring Cells Exchange Information): maps output of each cell to input of neighboring cells

Visualization of Model

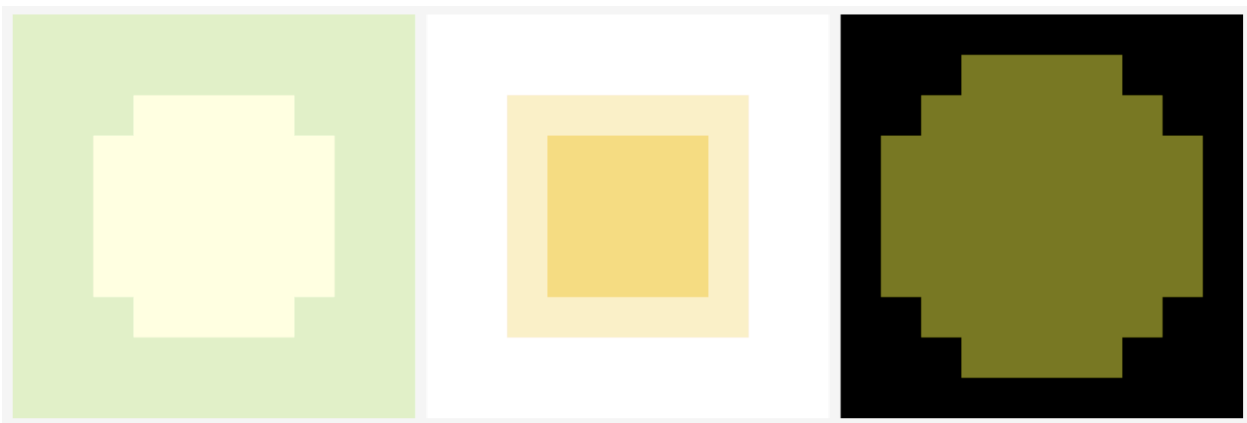

Figure 1: Cells-DEVS View of Nectar Foraging Simulation

The grid is visualized in three panels (left to right): nectar level, pollen level and number of bees.
In the Nectar Level panel:

- ⬜White to 🟩Green gradient
- Bins: 0-10, 10-25, 30-60, 60-80, 80-100

In the Pollen Level panel:

- ⬜White to 🟨Yellow to🟧Orange gradient
- Bins: 0-5, 5-10, 10-20, 20-30, 40-50

In the Number of Bees panel:

- ⬛Black to 🟨Yellow to 🟧Orange to🟥Red gradient
- Bins: 0-0.1, 0.1-3, 3-8, 8-15, 15-25, 25-50

Model Behaviour

The simulation implemented follows the ecological rules mentioned above within each discrete cell to model the relationship between foragers and resources. Each cell maintains a strict

carrying capacity to ensure that nectar and pollen grow at a steady rate until they reach their thresholds. Nectar level is capped at 100.0 units and pollen at 50.0 units.

Environment resources follow a regeneration-decay cycle. Nectar naturally grows at a base rate of 1.0 units per time step but also faces a 5% decay rate. One of the most important features of this model is that nectar regeneration is not a static constant but rather in a feedback loop with local pollen level. It will actually speed up if there is more pollen in the cell simulating how a well-learning environment is more productive. Similarly, the amount of bees present in the cell contributes to depletion of nectar.

The bee population has a maximum of 50 bees per cell to prevent overcrowding. These bees move the grid using a Von Neumann neighborhood. Rather than allowing diagonal dispersion, this constraint requires the bees migrate through North, East, South and West cells. Bee movement follows a resource-driven rule, where if the local nectar or pollen drop below a threshold of 8.0 units, 40% of the best choose to leave immediately. Additionally, if a neighbouring cell is 1.5 times better than their current cell, half of the bees will migrate towards the higher resource cell.

Additionally, the model had a feedback loop between bees and plants. When bees are in a cell, they consume nectar at a base rate of 0.3 per bee. Every time a bee visits a cell from a neighbour, it brings the 2.0 units of pollen with it. This incoming pollen then boosts the nectar regrowth rate, while departing bees carry some pollen away.

Experimental Frame

The goal of this section is to describe the testing strategies for the model under varying initial values. The tests below are used to observe how the model reacts to inputs that differ from the defined specifications. To view the .webm files, click on the test headings.

Test 1: Baseline Regeneration

In this test, all cells start with “bees = 0”. Since the cell update rule uses bee counts to drive nectar/pollen consumption , bee values remain zero throughout the simulation. The expected outcome is that nectar and pollen rise from their initial values and approach a stable level while remaining non-negative for the full simulation.

At t=0, the grid is homogenous where every cell starts with {nectar:10.0, pollen:5.0, bees: 0}. At t=1, every cell increments its resource level based on the growth rate (+1.0) minus the decay (0.05 * nectar level). By the end of simulation, t=50, resources approached their limits. The nectar stabilizes at 100.0 and pollen at 50.0. This test validates that resource constraints remain greater than or equal to zero and less than the threshold, and that regeneration behaves consistently.

Figure 2: Screenshot of the Simulation for Test1

Test 2: High Center Activity

This test initializes a small central zone of high activity; elevated bees, nectar, and pollen levels. It does this while keeping the rest of the grid at baseline. We expect that nectar and pollen in the high-activity region decrease rapidly due to increased bee consumption, and bee activity propagates outward to neighboring cells. The expected outcome is that the active region expands by the next time unit and that nectar initially active cells declines due to consumption exceeding regrowth.

At t=0, there is a zone of high resource and activity centered at [4,4] through [5,5] with {nectar:50.0, bees: 15} surrounded with cells that contain {nectar: 10.0, bees:0 }. At t=1, the bees begin to diffuse, as each cell keeps 50% of the bees of the cells and sends 2 bees to its orthogonal neighbours. In the center cells, consumption has outpaced regrowth. At t=10, the bees have spread as the nectar levels because the population density was highest there for the longest duration.

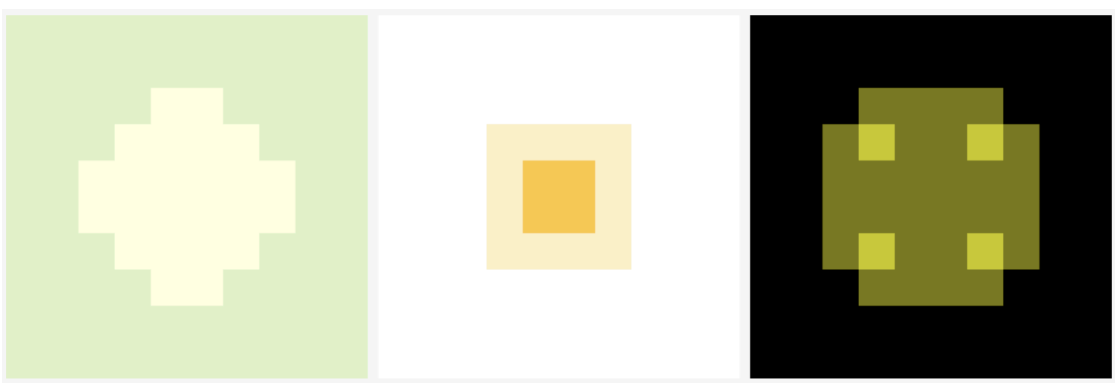

Figure 3: Screenshot of the Simulation for Test2

Test 3: Boundary Effects

The high activity zone is placed in a corner and reruns the same scenario under two settings: wrapped = true (normal boundary conditions) and wrapped = false (no neighbor wrap-around across edges. Wrap-around changes which cells are considered neighbors at the boundary, so we expect a difference in the rate of spread. The wrapped case should show equal or greater numbers of active cells by t=1 compared to the unwrapped case. In essence, we're testing that the boundary conditions affect emergent patterns in the grid.

When wrapped is set to false, at t=1, the bees at [0,0] only have two neighbours ([0,1] and [1,0]). The high population zone is against the boundaries, and restricted to a quarter-circle expanding towards the center. This shows physical barriers in a real environment that might constrain bee foraging zones.

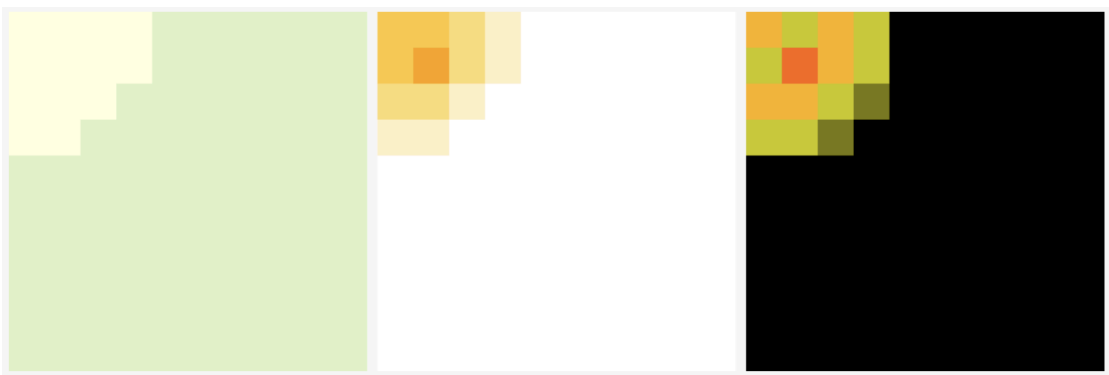

Figure 4: Screenshot of the Simulation for Test3

Test 4: Wrapped Corner

This simulation created 2x2 high activity zones in the Northwest corner (0,0) within a wrapped environment. The bee population does not encounter a boundary wall; instead, the expansion follows an orthogonal diamond pattern that appears simultaneously at all four corners of the grid.

When wrapped is set to true, at t=1, the bees at [0,0] have our neighbours: [0,1], [1,0], and the wrapped neighbors [0,9] and [9, 0]. As expected, activity appears instantly in all our corners of the grid. The single burst becomes our burst moving towards each other. This validates that the DEVSCoupled model correctly handles the coordinate wrapping.

The higher the nectar availability supports a more stable population of bees in the corner while the central population thins out which shows a resource driven migration. The bees gravitate toward areas where regeneration rate is met with their consumption rate.

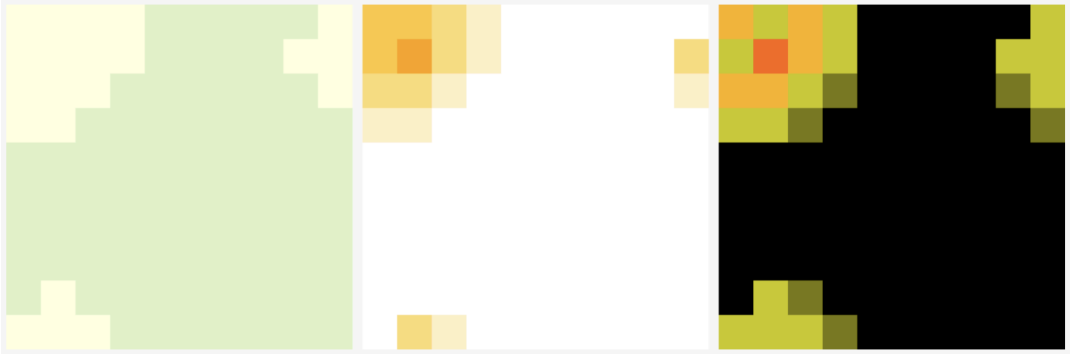

Figure 5: Screenshot of the Simulation for Test4

Test Summary

The four tests demonstrate distinct behaviors of the model under different conditions. Test 1 highlights the baseline, where the absence of bees results in nectar and pollen barely increasing due to the lack of pollinator participation. This confirms that the regrowth and decay mechanisms function correctly and that resource levels remain bounded.

Test 2 introduces a localized region of high activity, showing how increased bee density leads to rapid nectar depletion. As resources are consumed, bees begin to spread outward to neighboring cells, creating a wave of activity.

Test 3 examines boundary effects by placing the high-activity region at the edge of the grid without wrapping. The results show that bee movement is restricted due to fewer neighboring cells, causing the spread to be uneven and directed inward. This confirms that boundary conditions significantly influence how activity propagates across the grid.

Test 4 explores the impact of wrapping on the same corner scenario. With wrapping enabled, bees are no longer constrained by edges and can interact with cells across the grid boundaries. This results in activity appearing simultaneously in multiple corners and spreading more uniformly. Additionally, bees tend to stabilize in regions with higher nectar availability, demonstrating that movement is driven by resource conditions rather than uniform diffusion.

Together, these tests show that the model produces realistic emergent behavior driven by local interactions, resource availability, and grid topology.

Conclusion

The simulated Cell-DEVS model successfully demonstrated the interaction between foraging bee population and environment resources. Since the dynamics of the pollen and nectar were decoupled, the simulation showed a realistic ecosystem cycle by having continuous resource

regeneration and active depletion. Overall, this project validates the effectiveness of Cadmium modeling framework in simulating dynamic biological interactions.

References

[1] T. A. Revilla, “Plant–pollinator interaction model with separate pollen and nectar dynamics,” Journal of Theoretical Biology, vol. 606, p. 112096, Jun. 2025, doi: https://doi.org/10.1016/j.jtbi.2025.112096.